ENHANCED COMPUTATIONAL MODELING OF URBAN EVACUATION: A REVERSE DYNAMIC DIJKSTRA-BASED ROUTING AND FIRE PENALTY-BASED SIMULATION IN MESOSCOPIC CELLULAR AUTOMATA

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Abstract**—**High-density in urban environments face risks during fire-related disasters, often brought by inefficient evacuation strategies that lead to crowd congestions. This study enhances the mesoscopic cellular automata (MCA) framework by integrating a Reverse Dynamic Dijkstra (RDD) algorithm to optimize real-time pathfinding. Traditional models that rely on static, proximity-based assignments, the proposed model incorporates hazard-aware penalty fields that account for dynamic factors such as fire, smoke, debris, pedestrian, terrain difficulty, temporal obstruction, and density. Simulations were conducted on digitized university campus maps, specifically USTP CDO and USEP, comparing the enhanced model against baseline nearest exit and static Dijkstra configurations. Results demonstrate that while baseline models may perform well in hazard-neutral environments, the Reverse Dynamic Dijkstra implementation considerably outperforms them in complex penalty scenarios by rerouting calculated dijkstra paths. These findings provide a scalable and computationally efficient tool for urban disaster-preparedness and real-time evacuation management.

Keywords—Mesoscopic Cellular Automata(MCA), Reverse Dynamic Dijkstra(RDD), Pathfinding Optimization, Hazard-Aware Routing

I. Introduction

The dangers of fire especially in highly populated urban areas pose extreme risks, which led to evacuation strategies being essential in minimizing the casualties and damages [11], [19]. Although Cellular Automata (CA) models, especially the balanced Mesoscopic Cellular Automata (MCA), are useful tools for simulation, current evacuation strategies based on static, proximity-based exit allocation are insufficient. These models often lead to expected crowd congestion and stampedes at exits that are predominant [17], [9]. In order to solve this, we suggested improving the MCA framework by implementing a Reverse Dynamic Dijkstra algorithm. This method adapts exit allocation in real-time, optimizing routes based on exit distance, congestion, and penalty fields to simulate more realistic evacuee behavior. The goal is to improve evacuation safety and throughput compared to traditional MCA models.

A. Background of the Study

Rapid urbanization has escalated fire risks in high-density areas, where single ignitions can cause massive destruction[11], [16]. Significant incidents, such as the 2017 Manila fire, underscore the need for efficient planning [18]. While Cellular Automata(CA) models are essential simulation, microscopic versions are computationally expensive, while macroscopic models often oversimplify pedestrian behavior [17].

B. Problem Statement

The utility of Mesoscopic Cellular Automata(MCA) as an evacuation decision-support tool is hindered by two primary limitations.

- First, traditional models rely on proximity-based exit allocation which leads to disproportionate crowding at exits while alternatives remain underutilized [20], [7].
- Second, existing simulations often fail to fully integrate dynamic hazards-such as fire, debris, and smoke into pathfinding logic [7].

This results in impractical routing that ignores real-time environmental changes, such as reduced visibility and movement speeds. These gaps make current MCA models unreliable for large-scale environments where localized hazards and high pedestrian density can significantly dictate the outcome.

C. Objectives of the Study

The primary goal of this research is to enhance the MCA framework established in [7] to improve model flow and improve simulation realism. The specific objectives are:

1. Integrate Hazard-Aware Routing: Embed dynamic hazard penalties including fire, smoke and debris into the routing logic. By utilizing penalty based spread [10], the study evaluates how evolving environmental threats influence evacuation throughput and total clearance time.
2. Enhance evacuation routing through a Reverse Dynamic Dijkstra framework: Develop and implement an adaptive evacuation routing mechanism within the MCA framework by replacing the dynamic floor field with a Reverse Dynamic Dijkstra routing field enhanced by a congestion-penalty term. The Reverse Dijkstra formulation treats all exits as simultaneous sources, making it more suitable for multi-exit evacuation scenarios [23]. This routing enhancement aims to optimize path selection dynamically while accounting for congestion effects during evacuation.

D. Scope and Limitations of the Study

The modelling features included, and the conditions under which the evacuation simulations operate—as well as the inherent constraints and assumptions that may affect the generalizability and precision of the results.

E. Scope and Limitations

1. Scope

This study focuses on evacuation within university campuses using maps derived from institutional layouts. The research enhances the framework by [7], introducing the Reverse Dynamic Dijkstra (RDD) mechanism. Computing paths outwards from exits, the RDD approach minimizes redundant calculation and enables real-time rerouting as penalty hazards evolve. The model is integrated into a Mesoscopic Cellular Automata (MCA) framework and evaluated using performance metrics.

2. Limitations
   a. Behavioral Modeling: human behavior is simplified to route planning; psychological factors, panic, and physical injuries are not explicitly modeled[24].
   b. Fire Dynamics: Fire spread relies on probabilistic rules rather than fluid-dynamic physics to maintain computational efficiency [10].
   c. Environment: Simulations are restricted to ten university environments using segmented road cells rather than fine-grained grids [7].
   d. Generalizability: Findings are specific to the architectural layouts of the selected sites and may require adjustment for different urban contexts.
   e. Computational Hardware: Scale and fidelity are bound by grid resolution and available processing runtime.

F. Significance of the Study

This research provides three primary contributions to the field of disaster management:

1. Theoretical Advancement: It bridges graph-based pathfinding with cellular automata by extending the static Floor-Field approach [7]. By implementing a Reverse Dynamic Dijkstra (RDD) mechanism and a congestion-penalty formulation, it directly addresses the overcrowding and herding effects inherent in traditional models [5].
2. Algorithmic Efficiency: The framework leverages reverse-labeling search for multi-exit networks, significantly improving computational scalability for real-world environments compared to standard forward-Dijkstra implementations [9].
3. Practical Application: For university campuses and high-density facilities, the model serves as a decision-support tool for disaster-preparedness. The integration of congestion-aware logic provides actionable insights to minimize bottlenecks and reduce casualty risks during urban fire scenarios [8].

II. Review of Related Literature

A. Evacuation Models Classification

Evacuation models are categorized into three scales: microscopic, macroscopic, and mesoscopic. Microscopic models capture individual agent interactions but suffer from high computational costs in large-scale scenarios[4]. Macroscopic models treat crowds as continuous flows, gaining efficiency but losing local variety. Mesoscopic Cellular Automata (MCA) bridges these gaps by grouping evacuees into cells, maintaining spatial resolution and computational efficiency for large environments like university campuses[15], [17].

B. Hazard Dynamics and Pedestrian Behavior

Fire and smoke significantly degrade evacuation efficiency by reducing visibility and movement speed [6]. While CA models effectively simulate spatial outbreak [11], many traditional frameworks treat hazards as passive speed modifiers rather than active routing drivers. Recent studies emphasize that failing to couple fire progression with real-time congestion leads to underestimate evacuation times [5], [21].

C. Pathfinding and Congestion-Aware Routing

Shortest path algorithms, specifically Dijkstra, are standard for optimal routing. However, static implementations fail as environmental conditions evolve. Advanced frameworks such as Evacusafe integrate risk-based routing, penalizing high-density to prevent stampedes[8], [14]. Combining MCA with Reverse Dijktra fields allows for directional guidance at every cell, making the system expandable for multi-exit outdoor networks [9], [7].

D. Synthesis and Research Gap

Despite advancements, two critical gaps remain:

1. Proximity Bias: Most MCA models still rely on geometric distance for exit assignment, causing herding and bottlenecks at primary exits [7], [15].
2. Passive Hazard Integration: Existing mesoscopic models often ignore how the spread of penalties should actively reshape global route selection in real-time.

This study addresses these gaps by integrating a Reverse Dynamic Dijkstra (RDD) field with direct hazard-penalty formulations, enabling an expandable, hazard-aware simulation for complex urban environments.

III. Methodology

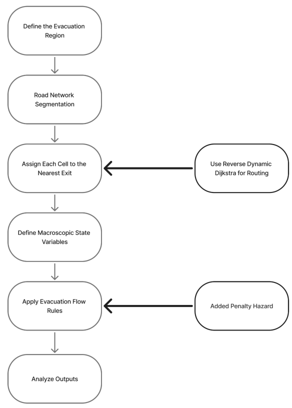
A. Methodology Overview

This study enhances the mesoscopic cellular automata (MCA) framework established by Lv et al. [1] for simulating pedestrian evacuation during urban fire scenarios. The proposed model integrates a Reverse Dynamic Dijkstra (RDD) algorithm for adaptive exit allocation and incorporates hazard-aware penalty fields to account for dynamic environmental conditions. All simulations are performed on mesoscopic road-cell grids derived from university campus maps, with results evaluated using standardized evacuation performance metrics.

B. Baseline Framework and Model Enhancements

The baseline MCA framework [1] consists of five sequential steps: (1) defining the evacuation region, (2) segmenting the road network into mesoscopic cells, (3) assigning each cell to the nearest exit, (4) defining macroscopic state variables, and (5) applying evacuation flow rules .

FIGURE I. MESOSCOPIC CELLULAR AUTOMATA FRAMEWORK WITH THE ADDED REVERSE DYNAMIC DIJKSTRA ROUTING AND PENALTY HAZARD.



This study modifies two components of this workflow shown on Figure I.:

- The "Assign Each Cell to the Nearest Exit" step is replaced with the Reverse Dynamic Dijkstra (RDD) routing mechanism, enabling cost-aware evacuation route computation rather than static nearest-exit assignments.
- The "Apply Evacuation Flow Rules" step is enhanced by incorporating hazard and congestion penalty values within each cell, allowing dynamic adjustment of traversal costs when fire or congestion conditions arise.

C. Enhanced MCA Pipeline

1) Evacuation Region Definition

The simulation domain is defined using digitized university campus maps. The University of Science and Technology of Southern Philippines Cagayan de Oro (USTP CDO) and University of Southeastern Philippines (USEP) campuses serve as primary test environments. These campuses are selected because their spatial layouts satisfy the mesoscopic mapping criteria described by Lv et al. [1]: connected outdoor roads, walkways, and open traversal areas that can be segmented into mesoscopic road cells. In accordance with mesoscopic principles, only traversable pathways (roads, footpaths, and walkable areas) are converted into simulation cells, while building interiors function solely as source-loading zones. Non-walkable regions including buildings, open fields, and parking lots are excluded from traversal.

2) Road Network Segmentation and Road Cell Construction

Following the definition of the evacuation region, campus pathways are converted into a mesoscopic road-cell network. Consistent with the mesoscopic principles of Lv et al. [1], each road cell represents a 10 m x 6 m road unit, providing sufficient spatial aggregation for crowd flow modeling while maintaining computational efficiency. The road centerlines are processed into structured road cells by identifying all traversable segments, campus roads, footpaths, and open movement areas. Each resulting road cell stores macroscopic variables including density, speed, and flow during simulation. Building-adjacent cells serve as source-loading cells, and exit-adjacent cells function as unloading nodes.

3) Routing Initialization Using Reverse Dynamic Dijkstra

In the baseline MCA model [1], each road cell is assigned to the nearest exit based solely on geometric distance, resulting in static exit choices that do not account for hazards or congestion. This study replaces the nearest-exit allocation with a Reverse Dynamic Dijkstra (RDD) approach. Instead of computing paths from every road cell to each exit, the RDD algorithm begins at the exits and propagates outward across the grid, assigning each cell a traversal cost and an initial preferred direction. This reverse expansion is computationally efficient, eliminating the need to compute thousands of source-to-exit paths individually [2]. During initialization, traversal costs reflect baseline factors such as road connectivity and geometric distance.

4) Cell State Update with Hazard and Congestion Penalties

At each timestep, the state of every road cell is updated using the mesoscopic CA flow equations defined by Lv et al. [1]. Each cell maintains its number of evacuees $N_i(t)$, density $\rho_i(t)$, speed $v_i(t)$, and flow $Q_i(t)$. These are updated using the standard MCA relationships:

$$\rho_i(t) = N_i(t) / A_i \tag{1}$$

$$v_i(t) = v_f * (1 - \rho_i(t) / \rho_{max}) \tag{2}$$

$$Q_i(t) = \rho_i(t) * v_i(t) \tag{3}$$

where i is the index of the road cell, t is the discrete simulation time step, $N_i(t)$ is the number of evacuees in cell i at time t, A_i is the physical area of road cell i (m^2), $\rho_i(t)$ is the pedestrian density in cell i at time t (persons/m^2), $v_i(t)$ is the mean walking speed in cell i at time t (m/s), v_f is the free-flow walking speed (1.3–1.5 m/s), and ρ_{max} is the maximum allowable pedestrian density (4-5 persons/m^2).

The enhanced model incorporates hazard and congestion effects by applying a penalty coefficient $C_{cell}(i,t)$ to the baseline MCA update. The penalty serves as a multiplicative reduction applied to the effective speed:

$$v_i^{eff}(t) = v_i(t) * (1 - C_{cell}(i,t)) \tag{4}$$

where $v_i^{eff}(t)$ is the effective speed after applying hazard and congestion penalties, representing the reduced movement capability caused by fire, smoke, debris, or local crowding.

5) Hazard and Congestion Penalty Formulation

To enable adaptive routing, each traversable cell is assigned a time-varying penalty value representing its current level of congestion and hazard exposure. The combined penalty for cell i at time t is expressed as:

$$C_{cell}(i,t) = \alpha*\rho^i(t) + \square*F_i(t) + \gamma*S_i(t) + \square*D_i(t) + \epsilon*T_i(r)_i + \zeta*T(o)_i \tag{5}$$

where ρ_i is the normalized pedestrian density (0-1), F_i is the fire intensity, S_i is the smoke density, D_i is the debris obstruction level, T_i(r) is the terrain difficulty modifier, T_i(o) is the temporary obstruction value, and alpha, beta, gamma, delta, epsilon, and zeta are tunable weight coefficients representing the relative influence of each factor.

The coefficient values are determined based on empirical studies: α = 0.6 for density influence [1], $\square$ = 1.0 for fire intensity [3], γ = 0.8 for smoke concentration [4], $\square$ = 0.4 for debris obstruction [5], ϵ = 0.2 for terrain difficulty [6], and ζ = 0.2 for temporary obstructions [5]. Penalty fields are updated every 5 seconds to balance responsiveness and computational efficiency. The composite penalty value is constrained by a maximum cap of 0.95, ensuring that agents are never fully immobilized even under extreme hazard accumulation.

This limitation preserves simulation continuity by preventing total movement stoppage while still representing severe mobility degradation. A rerouting threshold of 0.7 is adopted to regulate adaptive navigation behavior. This threshold enables agents to react promptly to hazardous conditions without triggering excessive or premature path recalculations. Setting a lower threshold (e.g., 0.6) would cause rerouting under relatively mild disturbances, potentially leading to unstable movement patterns and unnecessary computational overhead. For example, when crowd density (0.6) combines with even minor debris obstruction (0.2), the resulting penalty becomes $C_{cell} = 0.8$, Since this value exceeds the rerouting threshold, the agent initiates path reassessment, indicating that localized conditions have surpassed acceptable risk levels. The penalty value acts as a continuous speed modifier, where:

- Penalty = 0.0 indicate full movement speed
- Penalty = 0.95 indicate 5% of potential speed

This formulation models gradual movement degradation rather than binary interruption.

6) Dynamic Routing Update and Local Repair

The enhanced MCA framework continuously evaluates whether the precomputed routing field remains valid as conditions change. When hazards appear along previously optimal paths, or when congestion raises a cell's traversal cost, the model triggers a routing update. The repair procedure updates the RDD labels within the impacted region by performing a localized relaxation process, ensuring that only cells influenced by new penalty values are recalculated.

The time complexity of the initial reverse labeling is O((|V| + |E|) log |V|) using a heap-based priority queue, consistent with standard Dijkstra. Dynamic updates reuse the same data structures, allowing RDD to update only necessary cells rather than recomputing the entire field.

D. Experimental Design and Comparison Setup

1) Phase 1: Baseline vs. Enhanced MCA Without Hazard Penalties

Phase 1 compares the original Lv et al. [1] mesoscopic CA framework against the enhanced MCA under neutral conditions (without hazard penalties). Two configurations are evaluated:

- Baseline MCA - Nearest-Exit Routing: The control model follows the original mesoscopic CA setup with nearest-exit assignment and no RDD or hazard penalty.
- Enhanced MCA - Reverse Dynamic Dijkstra (No Penalties): Uses the full RDD framework with all penalty terms set to zero, isolating the benefits of the enhanced routing structure.

2) Phase 2: Hazard-Aware Routing Comparison

Phase 2 activates the full hazard-aware enhanced MCA framework. Two routing strategies are evaluated under identical hazard dynamics:

- Reverse Dynamic Dijkstra (RDD, Proposed): Computes a reverse cost-to-exit field and updates dynamically as penalty values change, with local label repair applied to affected regions.
- Baseline Routing with Hazard Penalty: Agents follow the original routing logic while incorporating the same hazard and congestion penalties, providing a controlled reference.

3) Simulation Scenarios

Both phases are tested on the mesoscopic road networks of USTP CDO and USEP campuses. Synthetic scenarios are generated by varying population agent size (1,000 - 10,000), stampede density (3.5 to 5.0 persons/m^2), and casualty rate parameters. Each scenario is run under all relevant configurations using identical initial maps, exits, and populations to ensure fair comparison. Results are averaged over multiple stochastic replications per scenario.

E. Evaluation Metric

The enhanced MCA model adopts eight evaluation metrics defined by Lv et al. [1] for consistent performance assessment:

TABLE I. EVACUATION PERFORMANCE METRICS

Metric	Description	Purpose
Exit Flow Rate	Evacuees exiting per second via each exit	Measure exit throughput
Remaining Evacuees	Sum of people still inside the network	Track evacuation progress
Density Evolution	Density in selected cells across time	Monitor congestion hotspots
Velocity	Mean walking speed from density	Assess movement efficiency
Exit Usage Distribution	Total evacuees per exit (accumulated)	Evaluate exit balance
Total Evacuation Time	Time until remaining < 1 person	Overall performance metric

F. Implementation

The enhanced MCA evacuation model is implemented as a modular system with four main layers: (1) Data Preparation Layer for QGIS-derived road-cell network preprocessing, (2) Routing Layer for RDD initialization and dynamic updates, (3) MCA Simulation Layer for applying mesoscopic state-update equations, and (4) Output and Analytics Layer for recording evacuation metrics and visualizations.

The model is developed using Python 3.7+ with supporting libraries including GeoPandas for spatial data handling, NetworkX for graph operations, NumPy for numerical computations, and Matplotlib for visualization. Simulations are executed on a modern CPU (quad-core or higher) with 8-16 GB RAM.

IV. RESULTS AND DISCUSSION

This section presents the results and discussion of the evacuation simulations conducted on the campus maps of University of Science and Technology of Southern Philippines (USTP) and University of Southeastern Philippines (USEP). The simulations compare two routing strategies: a Baseline approach representing standard shortest-path or default movement behavior, and the Dijkstra-based algorithm, which computes optimized shortest paths based on network weights.

A. Baseline and Dijkstra Performance

Table II. USTP Simulation Performance

Metrics	**Baseline**	**Dijkstra**	**Baseline with Penalty**	**Dijkstra with Penalty**
Exit Flow	16.041	15.260	8.669	15.239
Density	0.320	0.312	0.272	0.315
Velocity	0.873	0.881	1.049	0.874
Peak Density	25.857	25.955	25.366	25.702
Casualties	3024	3371	5417	3357
Evacuated	6976	6629	4583	6643

Evacuation Time	420.524	419.026	501.628	420.47
Exit 1	4617	4693	2640	4721
Exit 2	2359	1936	1943	1922

Table III. USEP Simulation Performance

Metrics	**Baseline**	**Dijkstra**	**Baseline with Penalty**	**Dijkstra with Penalty**
Exit Flow	13.574	13.744	8.0815	13.799
Density	0.507	0.504	0.400	0.503
Velocity	0.955	0.953	1.108	0.956
Peak Density	45.179	45.605	44.297	45.690
Casualties	4298	4224	5947	4203
Evacuated	5702	5776	4053	5797
Evacuation Time	404.848	404.94	471.226	404.85
Exit 1	963	1477	736	1483
Exit 2	1918	1472	1417	1494
Exit 3	2821	2827	1900	2820

Table II. and Table III. present the performance metrics of the evacuation simulations conducted on the USTP and USEP campus maps under four conditions: Baseline, Dijkstra, Baseline with hazard and fire penalties, and Dijkstra with hazard and fire penalties.

B. Discussions

The findings indicate that the integration of hazard and fire penalties significantly influences evacuation performance across both campus environments. Under non-penalty conditions, the differences between Baseline and Dijkstra simulations were minimal, suggesting that in static environments where risk factors are not considered, shortest-path optimization provides limited improvement over conventional routing behavior.

However, when hazard penalties were incorporated into the model, the Dijkstra algorithm demonstrated clear advantages. The significant reduction in casualties and evacuation time observed in both USTP and USEP simulations indicates that risk-aware routing enhances both efficiency and safety. By assigning higher traversal costs to hazardous areas, the Dijkstra model effectively redirected agents away from high-risk zones, resulting in improved evacuation outcomes.

Overall, the findings confirm that risk-aware path planning significantly enhances evacuation efficiency and safety outcomes. While traditional shortest-path routing provides modest benefits in static conditions, incorporating hazard-sensitive weights into the routing algorithm substantially improves system resilience in emergency scenarios. These results highlight the importance of adaptive evacuation planning in complex campus environments.

V. Conclusion

A. Conclusion

The primary objective of this research was to develop an enhanced MCA evacuation model that provides higher-fidelity representation of real-world fire scenarios. By integrating a Reverse Dynamic Dijkstra (RDD) algorithm and a multi-factor penalty formulation, this study has successfully addressed the limitations of static, proximity-based exit allocation.

Based on the experimental results, the following conclusions are drawn:

- Performance in Hazardous Conditions: The integrated Dijkstra routing has critically outperformed the baseline model in scenarios where hazards are present. While the baseline model often sends evacuees through dangerous areas simply because they are geographically closer, the enhanced model dynamically reroutes them towards safer paths.
- Scalability and Efficiency: The RDD approach proved more computationally efficient than standard forward Dijkstra for multi-exit environments. Propagating the costs outwards from exits, the model reduces redundant calculations and enables rapid, localized routing repairs when environmental conditions change.
- Impact of Agent Density: Simulations demonstrated that while baseline and enhanced models perform similarly at low agent counts(1,000 agents), the enhanced model dramatically reduces the casualties as population density increases (2,500-10,000 agents).

B. Recommendations

While the hazard-aware Reverse Dynamic Dijkstra (RDD) framework demonstrated measurable improvements in overall evacuation efficiency, the following areas are recommended for further investigation:

1. Controlled Environment Mapping: Future studies should consider adopting a manually constructed, noise-free spatial domain rather than relying on public datasets. This can allow researchers to isolate algorithmic performance from data irregularities and more precisely evaluate the interactions between penalty fields and congestion formation.
2. Advanced Penalty Modeling: Subsequent research should explore nonlinear weighting schemes and context-sensitive hazard prioritization within the MCA framework. Investigating adaptive scaling functions may yield further improvements in behavioral realism and stability under rapidly changing hazard conditions.

3. Cost-field Algorithm Variants: Further work should investigate alternative value-propagation or potential-field algorithms comparable to Dijkstra's method. Backward-labeling and reverse-search mechanisms beyond RDD provide additional computational advantages or behavioral refinements in multi-exit, dynamic environments.

Collectively, these directions will contribute to a deeper understanding of hazard-aware dynamics and enhance the scalability of mesoscopic evacuation models.

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